

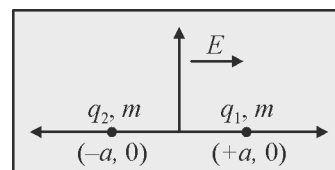
Illuminati - 2019

Advanced Physics Assignment -3B

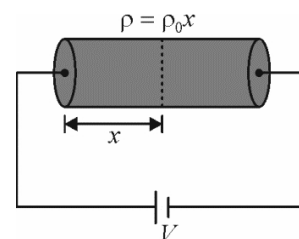
Electrostatics, DC-Circuits, Capacitors, Magnetism, EMI, AC-Circuits

SINGLE OPTION CORRECT TYPE

1. Two positively charged particles of charges q_1 and q_2 have mass m each. A uniform electric field having magnitude E exists in positive x direction as shown in the figure. The given two charged particles are released from rest at $t = 0$ as shown in the figure. If position of q_1 at $t = 2$ sec is given by coordinate $(+2a, 0)$ then the x -coordinate of q_2 at $t = 2$ sec is : (neglect gravitational interaction between the particles)

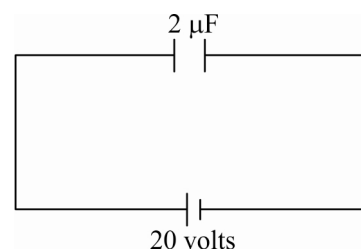


- (A) $\frac{q_1 + q_2}{m} E - 2a$ (B) $\frac{q_1 + q_2}{m} E - a$
 (C) $2\left(\frac{q_1 + q_2}{m}\right) E - 2a$ (D) $2\left(\frac{q_1 + q_2}{m}\right) E - a$
2. A cylindrical solid of length L and radius a is having varying resistivity given by $\rho = \rho_0 x$ where ρ_0 is a positive constant and x is measured from left end of solid. The cell shown in the figure is having emf V and negligible internal resistance. The electric field as a function of x is best described by:

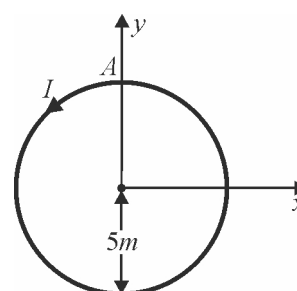


- ~~(A)~~ $\frac{2V}{L^2} x$ (B) $\frac{2V}{\rho_0 L^2} x$ (C) $\frac{V}{L^2} x$ (D) $\frac{V}{L}$

3. In the figure a capacitor of capacitance $2 \mu F$ is connected to a cell of emf 20 volt. The plates of the capacitor are drawn apart slowly to double the distance between them. The work done by the external agent on the plates is:



- (A) $-200 \mu J$ (B) $200 \mu J$
 (C) $400 \mu J$ (D) $-400 \mu J$
4. A ring of radius $5m$ is lying in the x - y plane and carrying current of 1A in anti-clockwise sense. If a uniform magnetic field $\vec{B} = 3\hat{i} + 4\hat{j}$ is switched on in space, then the co-ordinates of point about which the loop will have a tendency to lift up is :



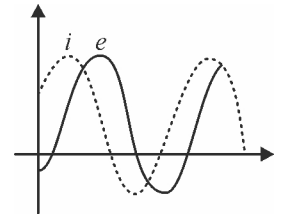
- (A) $(3, 4)$ (B) $(4, 3)$
 (C) $(3, 0)$ (D) $(0, 3)$

5. In a gravity free space, a smooth insulating ring of radius R , with a bead having charge q is placed horizontally in a uniform magnetic field of induction B_0 and perpendicular to the plane of ring. Starting from $t = 0$, the magnetic field is varying with time as $B(t) = B_0 + \alpha t$, where α is positive constant. The contact force between the ring and bead as a function of time is:

(A) $\frac{\alpha q^2 R t}{m} (2B_0 + \alpha t)$ (B) $\frac{\alpha q^2 R t}{4m} (2B_0 + \alpha t)$
 (C) $\frac{\alpha q^2 R t}{4m} (B_0 + \alpha t)$ (D) $\frac{4\alpha q^2 R t}{m} (2B_0 + \alpha t)$

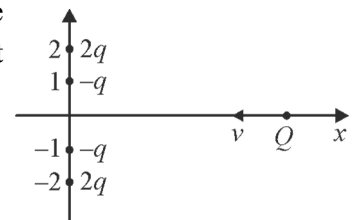
6. When an AC source of emf $e = E_0 \sin(100t)$ is connected across a circuit, the phase difference between emf e and current i in the circuit is observed to be $\pi/4$, as shown in figure. If the circuit consists possibly only of R-C or R-L or L-C in series, find the relationship between the two elements.

(A) $R = 1k\Omega$, $C = 100\mu F$ (B) $R = 1k\Omega$, $C = 1\mu F$
 (C) $R = 1k\Omega$, $L = 10H$ (D) $R = 1k\Omega$, $L = 1H$



7. Four charges are rigidly fixed along the Y axis as shown. A positive charge approaches the system along the X axis with initial speed just enough to cross the origin. Then its total energy at the origin is:

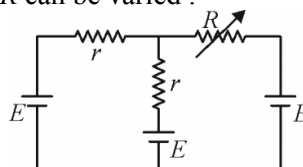
- (A) zero
 (B) positive
 (C) negative
 (D) data insufficient



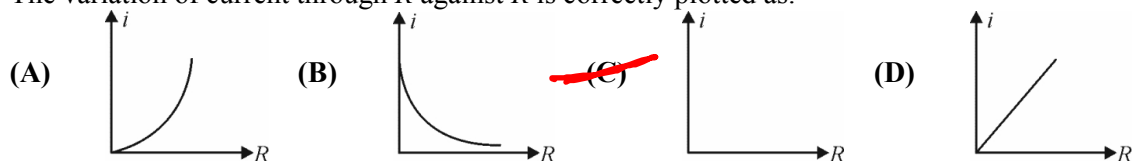
8. A dipole of dipole moment $\vec{P} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at point $A(2, -3, 1)$. The electric potential due to this dipole at the point $B(4, -1, 0)$ is equal to : (All the parameters specified here are in S.I. units)

(A) 2×10^9 volts (B) -2×10^9 volts (C) 3×10^9 volts (D) -3×10^9 volts

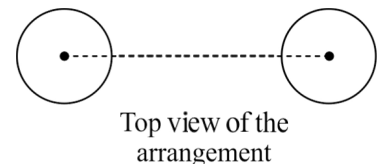
9. In the shown circuit the resistance R can be varied :



The variation of current through R against R is correctly plotted as:

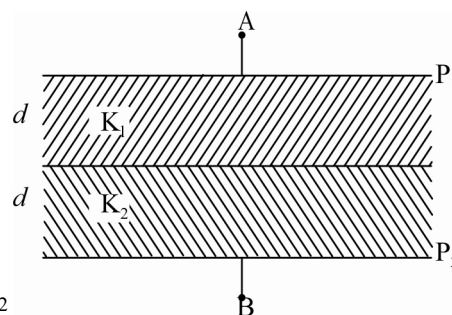


10. In the figure shown two long straight wires with the same cross-sections are arranged in air, parallel to one another. The distance between the axis of the wire is η times larger than the radius of wire's cross section. Capacitance of the wires per unit length would be (Taken $\eta \gg 1$)



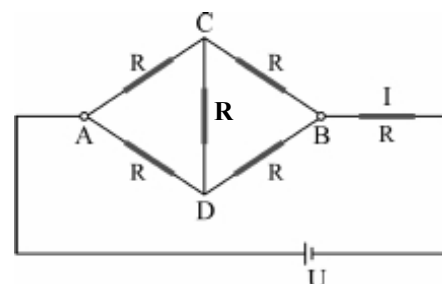
(A) $\frac{2\pi\epsilon_0}{\ln \eta}$ (B) $\frac{\pi\epsilon_0}{2\ln \eta}$
 (C) $\frac{\pi\epsilon_0}{\ln \eta}$ (D) Information insufficient

11. In the figure shown P_1 and P_2 are two conducting plates having charges of equal magnitude and opposite sign. Two dielectrics of dielectric constant K_1 and K_2 fill the space between the plates as shown in the figure. The ratio of electrical energy in 1st dielectric to that in the 2nd dielectric is:



- (A) 1:1 (B) $K_1 : K_2$
(C) $K_2 : K_1$ (D) $K_2^2 : K_1^2$

12. If the values of the resistance of the resistors shown in the figure are equal then the current in the main branch is I. By what factor does this current change if the resistance of the two resistors, which are diagonally opposite each other, is doubled?

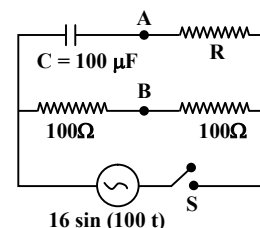


- (A) 1:1 (B) 2:3
(C) 6:5 (D) 5:6

13. A non-conducting sphere of radius R charged uniformly with surface density σ rotates with an angular velocity ω about a diametrical axis passing through its centre. Find the magnetic induction due to the rotating charge at the centre of the sphere.

- (A) $\frac{2}{3}\mu_0\sigma\omega R$ (B) $\mu_0\sigma\omega R$ (C) $\frac{1}{3}\mu_0\sigma\omega R$ (D) $\frac{\mu_0\sigma\omega R}{2}$

14. An uncharged capacitor $C = 100 \mu\text{F}$ with a resistor R is connected with AC source as shown in the figure. If R is 50Ω and switch S is closed at $t = 0$, the maximum value of $(V_A - V_B)$ in volts is.

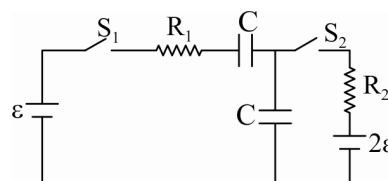


- (A) 2
(B) 4
(C) 6
(D) 8

15. A coil, a capacitor and AC source of rms voltage 24 V and connected in series. By varying the frequency of the source, a maximum rms current of 6A is observed. If coil is connected to a battery of emf 12V and internal resistance 4Ω , then current through it in steady state is :

- (A) 2.4 A (B) 1.8 A (C) 1.5 A (D) 1.2 A

16. In the circuit shown, switch S_2 is closed first and is kept closed for a long time. Now S_1 is closed. Just after that instant the current through S_1 is:

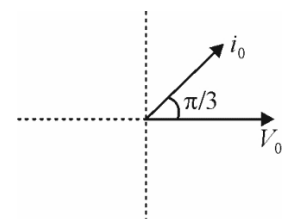
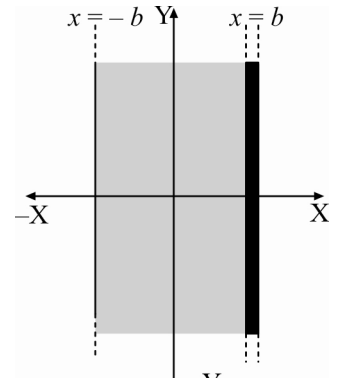


- (A) $\frac{\varepsilon}{R_1}$ towards right
(B) $\frac{\varepsilon}{R_1}$ towards left
(C) Zero
(D) $\frac{2\varepsilon}{R_1}$ towards right

17. When $100V$ DC is applied across a solenoid, a current of 1 A flows in it. When $100V$ AC is applied across the same coil, the current drops to $0.5A$. If the frequency of the AC source is 50 Hz , the impedance and inductance of solenoid are :
- (A) $100\Omega, 0.93H$ (B) $200\Omega, 1.0H$ (C) $10\Omega, 0.86H$ (D) $200\Omega, 0.55H$
18. Two ammeters, 1 and 2, have different internal resistances : r_1 (known) and r_2 (unknown). Each ammeter has an analog scale such that the angular deviation of the needle from zero is proportional to the current. Initially, the ammeters are connected in series and then to a source of constant voltage. The deviations of the needles of the ammeters are θ_1 and θ_2 , respectively. The ammeters are then connected in parallel and then to the same voltage source. This time, the deviations of the needles are θ'_1 and θ'_2 respectively. Find r_2 .
- (A) $r_2 = r_1 \frac{\theta_1 \theta_2}{\theta'_1 \theta'_2}$ (B) $r_2 = r_1 \frac{\theta_1 \theta'_1}{\theta_2 \theta'_2}$ (C) $r_2 = r_1 \frac{\theta_2 \theta'_1}{\theta_1 \theta'_2}$ (D) $r_2 = r_1 \frac{\theta_1 \theta'_2}{\theta_2 \theta'_1}$

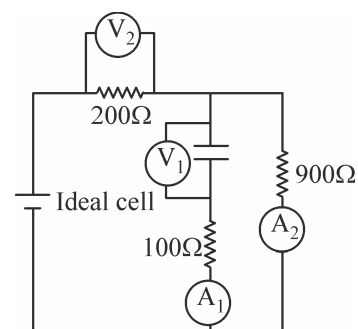
ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

19. The figure contains an infinite slab having current per unit area of $\vec{J}_1 = J\hat{k}$ between the infinite planes at $x = -b$ and $x = b$. Slightly to the right of $x = b$, an infinite thin sheet is kept. It carries a current per unit length $\vec{k}_2 = 2bJ(-\hat{k})$. Choose the correct option(s):
- (A) The field due to sheet at any point is $\mu_0 Jb$
- (B) The field due to sheet at any point is $\frac{\mu_0 Jb}{2}$
- (C) The magnetic field due to slab at a point inside the slab must be independent of y co-ordinate and y component of field must be a function of x
- (D) Magnetic field at a point (x, y) inside slab, due to slab is $\mu_0 Jx$
20. A particle of specific charge ' α ' is projected from origin at $t = 0$ with a velocity $\vec{V} = V_0(\hat{i} + \hat{k})$ in a magnetic field $\vec{B} = -B_0\hat{k}$. Then : (Mass of particle = 1 unit)
- (A) At $t = \frac{\pi}{\alpha B_0}$, velocity of the particle is $-V_0(\hat{i} - \hat{k})$
- (B) At $t = \frac{\pi}{4\alpha B_0}$, speed of the particle is V_0
- (C) At $t = \frac{2\pi}{\alpha B_0}$, magnitude of displacement of the particle is more than $\frac{2V_0}{\alpha B_0}$
- (D) At $t = \frac{2\pi}{\alpha B_0}$, distance travelled by the particle is less than $\frac{2\sqrt{2}\pi V_0}{\alpha B_0}$
21. For a RLC series circuit, phasors of current i and applied voltage $V = V_0 \sin \omega t$ are shown in diagram at $t = 0$. Which of the following is/are correct?
- (A) At $t = \frac{\pi}{2\omega}$, instantaneous power supplied by source is negative
- (B) From $0 < t < \frac{2\pi}{3\omega}$, average power supplied by source is positive
- (C) At $t = \frac{5\pi}{6\omega}$, instantaneous power supplied by source is negative
- (D) If ω is increased slightly, angle between the two phasors decreases.



22. An electrical circuit is shown in the given figure. The resistance of each voltmeter is infinite and each ammeter is $100\ \Omega$. The charge on the capacitor of $100\ \mu\text{F}$ in steady state is 4mC . Choose correct statements regarding above circuit.

- (A) Reading of voltmeter V_2 is 16V
 (B) Reading of ammeter A_1 is zero and A_2 is $1/25\text{ A}$
 (C) Reading of voltmeter V_1 is 40V
 (D) EMF of the ideal cell is 66V



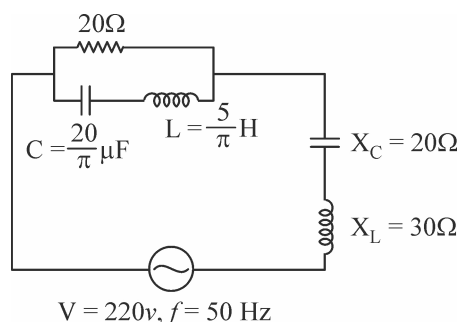
23. Two concentric, coplanar, circular loop of wire, with different diameter carry current in the same sense as shown in the figure. Which of the following statement(s) is(are) correct ?



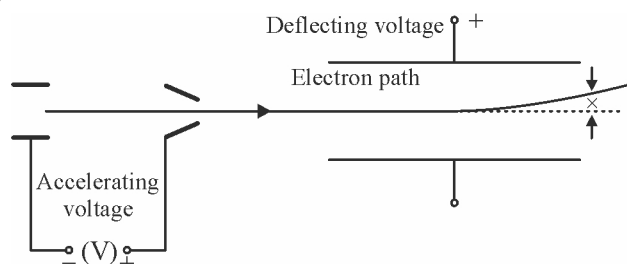
- (A) the magnetic force exerted by the outer loop on a short portion of the inner loop is radially outward
 (B) the magnetic force exerted by the outer loop on a short portion of the inner loop is radially inward
 (C) the net magnetic force exerted by the outer loop on a whole inner loop is non-zero and is radially outward
 (D) the net magnetic force exerted by the outer loop on a whole inner loop is zero

24. The circuit shown below is :

- (A) purely resistive
 (B) purely inductive
 (C) purely capacitive
 (D) zero power factor



25. The diagram shows part of an evacuated tube in which a stream of electrons from an electron gun passes between a pair of parallel large deflecting plates. The vertical displacement of the electron beam as it leaves the parallel plates is x , (Do not consider gravity and the electron enters the deflecting region parallel to the plates)



Which one of A to D below will change the displacement x of the beam as it leaves the parallel plates.

- (A) increasing the accelerating voltage
 (B) increasing the deflecting voltage
 (C) increasing the distance between the electron gun and the deflecting plates
 (D) increasing the distance between the two deflecting plates

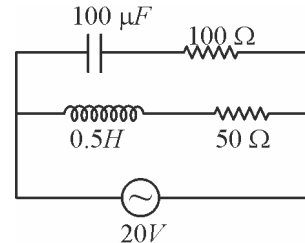
26. A capacitor (without dielectric) is discharging through a resistor. At some instant a dielectric is inserted between the plates, then:

(A) Just after the insertion of the dielectric, current will decrease
 (B) Just after the insertion of the dielectric, charge on capacitor will increase
 (C) Just after the insertion of the dielectric, energy stored in the capacitor will decrease
 (D) After the insertion of the dielectric, time constant will increase

27. For the circuit shown in figure, the AC source has $\omega = 100 \text{ rad s}^{-1}$. Considering the inductor and capacitor to be ideal.

Correct alternative are :

(A) the current through the circuit I is 0.3 A
 (B) the current through the circuit I is $0.3\sqrt{2} \text{ A}$
 (C) the voltage across 100Ω resistor is $10\sqrt{2} \text{ V}$
 (D) the voltage across 50Ω resistor is 10 V

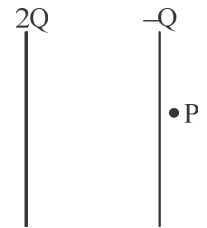


28. In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is ' C '. P is a point outside the capacitor and close to the plate of charge $-Q$. The distance between the plates is ' d ', select correct alternatives:

(A) A point charge at point ' P ' will experience electric force due to capacitor
 (B) The potential difference between the plates will be $\frac{3Q}{2C}$

(C) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$

(D) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\epsilon_0 d^2}$



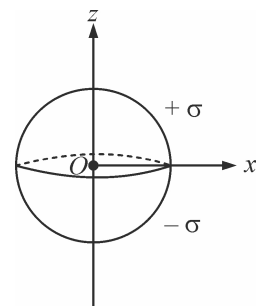
29. A hollow, insulating spherical shell has a surface charge distribution placed upon it, such that the upper hemisphere has a uniform surface charge density $+\sigma$, while the lower hemisphere has a uniform surface charge density $-\sigma$, as given in the figure.

(A) The field at all points of x - y plane within the sphere points in the $-ve z$ direction

(B) All points of the x - y plane within the sphere are equipotential

(C) The field at all points on z -axis which are on either side of origin outside the sphere is in opposite direction

(D) The field at points on z -axis which are on either side of origin inside the sphere are in opposite direction



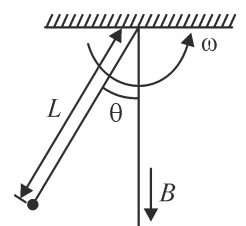
30. A charged particle having its charge to mass ratio as β goes in a conical pendulum of length L making an angle θ with vertical and angular velocity ω . If a magnetic field B is directed vertically downwards (see figure), then choose correct options.

(A) $B = \frac{1}{\beta} \left[\omega - \frac{g}{\omega L \cos \theta} \right]$

(B) Angular momentum of the particle about the point of suspension remains constant

(C) If the direction of B were reversed maintaining same ω and L , then θ will remain unchanged

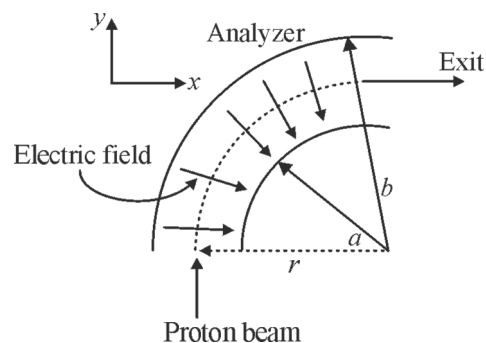
(D) Rate of change of angular momentum of the particle about the point of suspension is not a constant vector.



PARAGRAPH TYPE

Paragraph for Questions 31 - 33

Figure shows a schematic view of an electrostatic analyzer. It can sort out charged particles by speed and charge to mass ratio. Spacecraft use such analyzers to characterize charged particles in interplanetary space. Two curved metal plates establish an electric field given by $E = E_0 \left(\frac{b}{r} \right)$ where E_0 and b are positive constants with unit of electric field and length. The field points toward the centre of curvature and r is distance from centre. There is no influence of gravity. Proton (charge $+e$ mass ' m ') enters along y -axis and exits along x -axis while moving along a circular path.



31. Speed with which proton is to be projected ' v ' and centripetal acceleration of electron ' a_c ' is given by respectively :

(A) $v = \sqrt{\frac{eE_0b}{m}}; a_c = \frac{2e}{m} E_0 \left(\frac{b}{r} \right)$ (B) $v = \sqrt{\frac{2eE_0b}{m}}; a_c = \frac{e}{2m} E_0 \left(\frac{b}{r} \right)$

(C) $v = \sqrt{\frac{eE_0b}{2m}}; a_c = \frac{2e}{m} E_0 \left(\frac{b}{r} \right)$ ~~(D)~~ $v = \sqrt{\frac{eE_0b}{m}}; a_c = \frac{e}{m} E_0 \left(\frac{b}{r} \right)$

32. Mark the **INCORRECT** option :

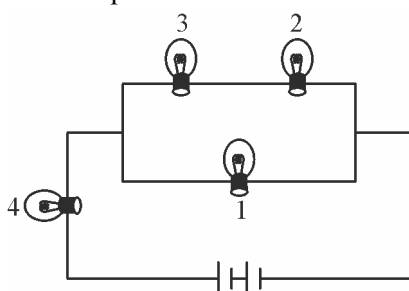
- (A) Work done by electric field on proton is zero
- (B) If $v = \sqrt{\frac{2eE_0b}{m}}$ proton may strike outer surface of analyzer
- (C) If $v = \sqrt{\frac{2eE_0b}{m}}$ proton may strike inner surface of analyzer
- (D) If an electron is released with zero initial velocity from inner surface of analyzer it will strike outer surface with velocity $v = \sqrt{\frac{2eE_0b}{m_e} \ln \left(\frac{b}{a} \right)}$, where m_e is mass of electron

33. Mark the correct option.

- (A) If E_0 is made larger then in order to maintain same trajectory, initial speed has to be decreased
- (B) If proton enters closer to the inner surface it will require smaller speed to follow circular trajectory
- (C) It does not matter where the protons enter the device it requires same speed to follow circular trajectory
- (D) A deuteron (charge $+e$, mass $2m$) will require greater speed as compared to proton to follow circular trajectory

Paragraph for Questions 34 - 36

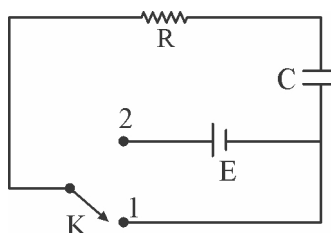
All bulbs consume same power. The resistance of bulb 1 is 36Ω .



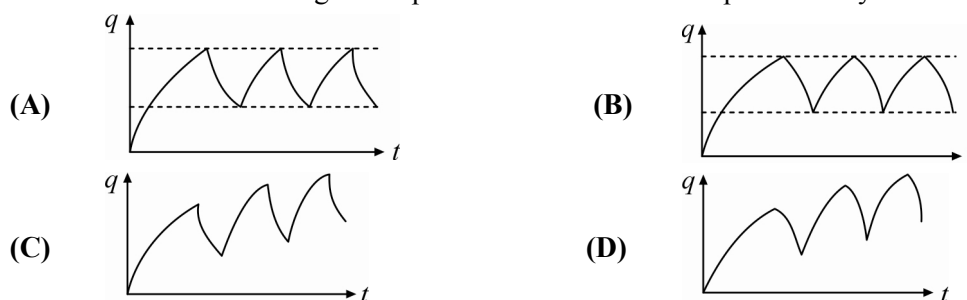
34. What is the resistance of bulb 3 ?
 (A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω
35. What is the resistance of bulb 4 ?
 (A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω
36. What is the voltage output of the battery if the power of each bulb is 4 W ?
 (A) 12 V (B) 16 V (C) 24 V (D) None of these

Paragraph for Questions 37 - 39

In the shown circuit involving a resistor of resistance $R\Omega$, capacitor of capacitance C farad and an ideal cell of emf E volts, the capacitor is initially uncharged and the key is in position 1. At $t = 0$ second, the key is pushed to position 2 for $t_0 = RC$ seconds and then key is pushed back to position 1 for $t_0 = RC$ seconds. This process is repeated again and again. Assume the time taken to push key from position 1 to 2 and vice versa to be negligible.

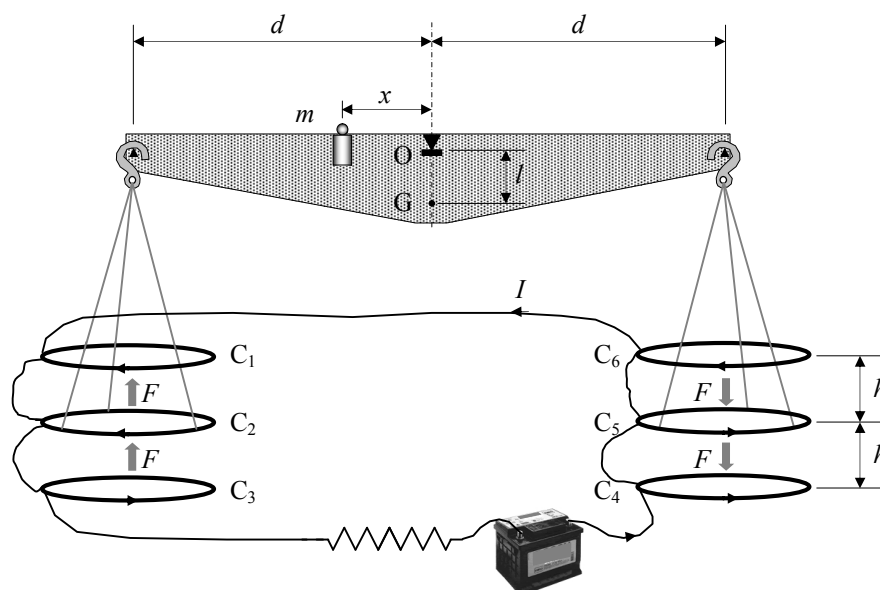


37. The charge on capacitor at $t = 2RC$ second is :
 (A) CE (B) $CE\left(1 - \frac{1}{e}\right)$ (C) $CE\left(\frac{1}{e} - \frac{1}{e^2}\right)$ (D) $CE\left(1 - \frac{1}{e} + \frac{1}{e^2}\right)$
38. The current through the resistance at $t = 1.5RC$ seconds is :
 (A) $\frac{E}{e^2 R}\left(1 - \frac{1}{e}\right)$ (B) $\frac{E}{eR}\left(1 - \frac{1}{e}\right)$ (C) $\frac{E}{R}\left(1 - \frac{1}{e}\right)$ (D) $\frac{E}{\sqrt{e}R}\left(1 - \frac{1}{e}\right)$
39. Then the variation of charge on capacitor with time is best represented by :



Paragraph for Questions 40 - 42**Determination of the ampere**

Passing a current through two conductors and measuring the force between them provides an absolute determination of the current itself. The “Current Balance” designed by Lord Kelvin in 1882 exploits this method. It consists of six identical single turn coils $C_1 \dots C_6$ of radius a , connected in series. As shown in figure, the fixed coils C_1, C_3, C_4 , and C_6 are on two horizontal planes separated by a small distance $2h$. The coils C_2 and C_5 are carried on balance arms of length d , and they are, in equilibrium, equidistant from both planes. The current I flows through the various coils in such a direction that the magnetic force on C_2 is upwards while that on C_5 is downwards. A mass m at a distance x from the fulcrum O is required to restore the balance to the equilibrium position described above when the current flows through the circuit. Let M be the mass of the balance (except for m and the hanging parts), G its centre of mass and l the distance $\overline{OG}$.



40. Compute the force F on C_2 due to the magnetic interaction with C_1 . For simplicity assume that the force per unit length is the one corresponding to two long, straight wires carrying parallel currents.
- (A) $\frac{\mu_0 I^2}{2\pi h}$ (B) $\frac{\mu_0 I^2 a}{h}$ (C) $\frac{2\mu_0 I^2 a}{h}$ (D) $\frac{\mu_0 I^2}{ah}$
41. The current I is measured when the balance is in equilibrium. Give the value of I in terms of the physical parameters of the system. The dimensions of the apparatus are such that we can neglect the mutual effects of the coils on the left and on the right.
- (A) $\left(\frac{mghx}{4\mu_0 ad}\right)^{1/2}$ (B) $\left(\frac{mghx}{8\mu_0 ad}\right)^{1/2}$ (C) $\left(\frac{mghx}{2\mu_0 ad}\right)^{1/2}$ (D) $\left(\frac{mga x}{4\mu_0 hd}\right)^{1/2}$
42. The balance equilibrium is stable against deviations producing small changes δ_z in the height of C_2 and $-\delta_z$ in C_5 . Compute the maximum value $\delta_{z_{\max}}$ so that the balance still returns towards the equilibrium position when it is released. (Consider that the coils centres remain approximately aligned)
- (A) $\frac{M\ell h^2}{2mxd}$ (B) $\frac{M\ell h^2}{mxd}$ (C) $\frac{M\ell h}{mxd}$ (D) $\frac{M\ell h}{mx}$

Paragraph for Questions 43 - 45**Method of images**

A point charge q is placed in the vicinity of a grounded metallic sphere of radius R [see figure (i)], and consequently a surface charge distribution is induced on the sphere. To calculate the electric field and potential from the distribution of the surface charge is a formidable task. However, the calculation can be considerably simplified by using the so called method of images. In this method, the electric field and potential produced by the charge distributed on the sphere can be represented as an electric field and potential of a single point charge q' placed inside the sphere (you do not have to prove it). Note : The electric field of this image charge q' reproduces the electric field and the potential only outside the sphere (including its surface).

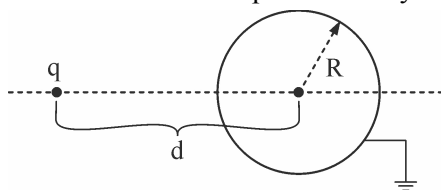


Figure (i)

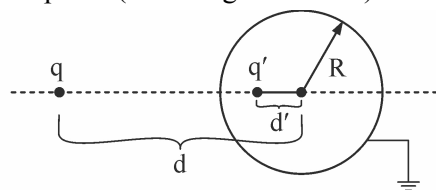


Figure (ii)

The symmetry of the problem dictates that the charge q' should be placed on the line connecting the point charge q and the centre of the sphere [see figure (ii)]. The value of q' and its distance from the centre of sphere d' can be found by using the fact that potential at all points of the sphere must be zero.

43. Value of q' is :

- ~~(A)~~ $q \frac{R}{d}$ (B) $q \frac{d}{R}$ (C) $-q \frac{R}{d}$ (D) $-q$

44. Value of d' is :

- (A) d ~~(B)~~ $\frac{R^2}{d}$ (C) $\frac{d^2}{R}$ (D) $\frac{2R^2}{d}$

45. Find the magnitude of force acting on charge q .

- ~~(A)~~ $\frac{1}{4\pi\epsilon_0} \frac{q^2 R d}{(d^2 - R^2)^2}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{(d^2 - R^2)}$
 (C) $\frac{1}{4\pi\epsilon_0} \frac{q^2 d^2}{(d^2 - R^2)^2}$ (D) zero

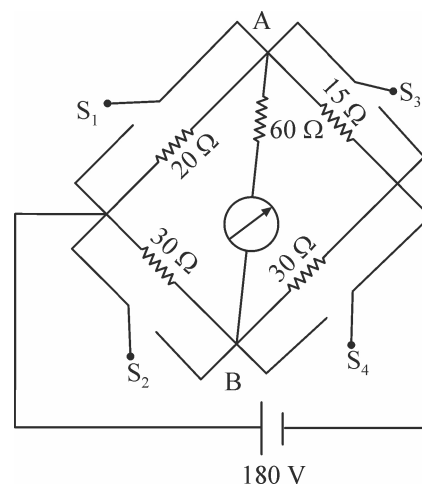
MATCH THE COLUMN TYPE

46. A particle enters a space where exists uniform magnetic field $\vec{B} = B_x \vec{i} + B_y \vec{j} + B_z \vec{k}$ and uniform electric field $\vec{E} = E_x \vec{i} + E_y \vec{j} + E_z \vec{k}$ with initial velocity $\vec{u} = u_x \vec{i} + u_y \vec{j} + u_z \vec{k}$. Depending on the values of various components the particle selects a path. Match the entries of column A with the entries of column B. The component other than specified in column A in each entry are non-zero. Neglect gravity.

Column I**Column II**

- | | |
|--|---|
| (A) $B_y = B_z = E_x = E_z = 0 ; u = 0$ | (p) circle |
| (B) $E = 0, u_x B_x + u_y B_y \neq -u_z B_z$ | (q) helix with uniform pitch and constant radius |
| (C) $\vec{u} \times \vec{B} = 0, \vec{u} \times \vec{E} = 0$ | (r) cycloid |
| (D) $\vec{u} \perp \vec{B}, \vec{B} \parallel \vec{E}$ | (s) helix with variable pitch and constant radius |
| | (t) straight line |

47. Consider the circuit shown. The resistance connected between the junction A and B is $60\ \Omega$ including the resistance of the galvanometer. The switches have no resistance when shorted and infinite resistance when opened. All the switches are initially open and they are closed as given in column I. Match the condition in column I with the direction of current through galvanometer and the value of the current through the battery in column II.

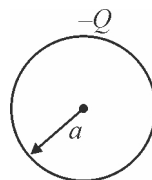


- | Column I | Column II |
|---------------------------------|--|
| (A) Only switch S_1 is closed | (p) Current from A to B |
| (B) Only switch S_2 is closed | (q) Current from B to A |
| (C) only switch S_3 is closed | (r) Current through the battery is 12.0 A |
| (D) Only switch S_4 is closed | (s) Current through the battery is 15.6 A |
| | (t) Current through the Galvanometer is 1.2 A |
48. In each situation of column -I, some charge distributions are given with all details explained. The electrostatic potential energy and its nature is given in column -II. Then match situation in column -I with the corresponding results in column -II and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the OMR.

Column-I

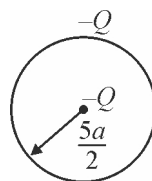
Column-II

- 85 (A) A thin shell of radius a and having a charge $-Q$ uniformly distributed over its surface as shown



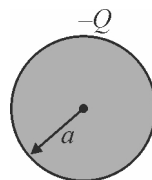
(p) $\frac{1}{8\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

- 95 (B) A thin shell of radius $\frac{5a}{2}$ and having a charge $-Q$ uniformly distributed over its surface and a point charge $-Q$ placed at its centre as shown



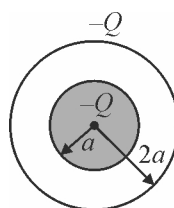
(q) $\frac{3}{20\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

- 95 (C) A solid sphere of radius a and having a charge $-Q$ uniformly distributed throughout its volume as shown.



(r) $\frac{2}{5\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

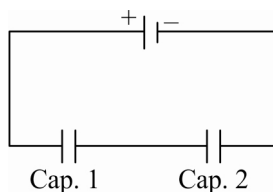
- 5 (D) A solid sphere of radius a and having a charge $-Q$ uniformly distributed throughout its volume. The solid sphere is surrounded by a concentric thin uniformly charged spherical shell of radius $2a$ and carrying charge $-Q$ as shown



(s) Positive in sign

(t) Zero

49. Two identical capacitors are connected in series, and the combination is connected with a battery as shown. Some changes in the capacitor 1 are now made independently after the steady state is achieved, listed in column-I. Some effects which may occur in new steady state due to these changes on the capacitor 2 are listed in column-II. Match the changes on capacitor 1 in column-I with corresponding effect on capacitor 2 in column-II.

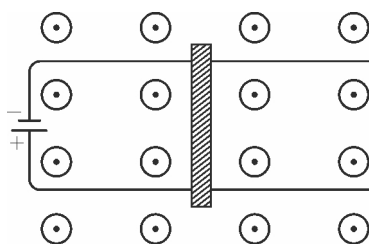
**Column-I**

- (A) A dielectric slab is inserted
 (B) Separation between plates increased
 (C) A metal plate is inserted connecting both plates
 (D) The left plate is grounded

Column-II

- (p) Charge on the capacitor increases
 (q) Charge on the capacitor decreases
 (r) Energy stored in capacitor increases
 (s) Energy stored in capacitor decreases
 (t) No change occurs

50. A conducting rod (mass m , resistance R) rests on two frictionless and resistanceless rails distant l apart in a uniform magnetic field B . At $t = 0$, the rod is at rest and a source of emf E_0 is connected to points a and b as shown below.



Now, match the following columns and select the correct option from the codes given below.

Column I**Column II**

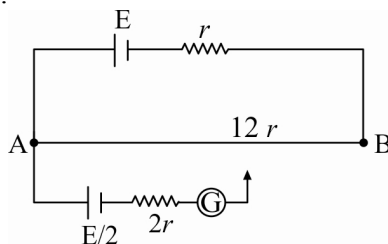
- | | |
|--|---|
| I. Speed of rod at time t when source gives a constant current I | (p) ∞ |
| II. Speed of rod at time t when source gives a constant emf E_0 | (q) $\frac{E_0}{Bl}$ |
| III. Terminal speed of rod when source gives a constant current I | (r) $\frac{E_0}{Bl} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right)$ |
| IV. Terminal speed of rod when source gives a constant emf E_0 | (s) $\frac{E_0 l B t}{mR}$ |

Codes :

- | | I | II | III | IV | | I | II | III | IV |
|-----|---|----|-----|----|-----|---|----|-----|----|
| (A) | p | q | r | s | (B) | q | r | s | p |
| (C) | r | s | p | q | (D) | s | r | p | q |

NUMERICAL/SUBJECTIVE TYPE

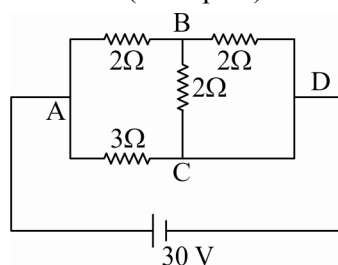
51. Consider a cube of side $a = 0.1 \text{ m}$ placed such that its six faces are given by equations $x = 0, x = +a, y = 0, y = +a, z = 0$ and $z = +a$, placed in electric field given by $\vec{E} = x^2 \hat{i} + y \hat{j} \text{ N/C}$. Find the electric flux crossing out of the cube in the unit of $10^{-4} \text{ Nm}^2/\text{C}$.
52. Consider the potentiometer circuit arranged as in figure. The potentiometer wire AB is 300 cm long. If the jockey touches the wire at a distance of 275 cm from A , then $\left(\frac{3E}{Nr}\right)$ current flow through galvanometer find value of N .



53. A parallel plate capacitor is to be designed which is to be connected across 1 kV potential difference. The dielectric material which is to be filled between the plates has dielectric constant $K = 6\pi$ and dielectric strength 10^7 V/m . For safety, the electric field is never to exceed 10% of the dielectric strength. With such specifications, if we want a capacitor of capacitance 50 pF, what minimum area (in mm^2) of plates is required for safe working?

(use $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \text{ in MKS}$)

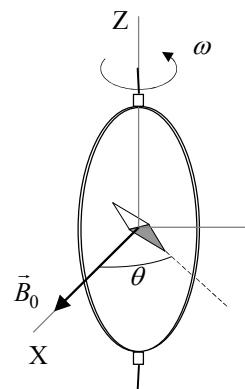
54. Find current in the branch CD of the circuit (in ampere).



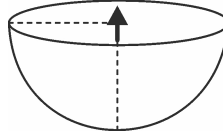
55. A closed circular coil of N turns, radius a and total resistance R is rotated with uniform angular velocity ω about a vertical diameter in a horizontal magnetic field $\vec{B}_0 = B_0 \hat{i}$.

(i) Compute the electromotive force ε induced in the coil, and also the mean power $\langle P \rangle$ required for maintaining the coil in motion. Neglect the coil self inductance.

(ii) A small magnetic needle is placed at the center of the coil, as shown in figure. It is free to turn slowly around the Z axis in a horizontal plane, but it cannot follow the rapid rotation of the coil. Once the stationary regime is reached, the needle will set at a direction making a small angle θ with $\vec{B}_0$. Compute the resistance R of the coil in terms of this angle and the other parameters of the system.



56. A small freely oriented electric dipole of dipole moment $\vec{P}$ is placed at the centre of a charged hemispherical cup of surface charge density σ coul/m² with dipole moment vector along axis of cup as shown in figure. Find the period of small oscillations of dipole about diametrical axis of base of cup. Moment of inertia of dipole about the given axis is I .

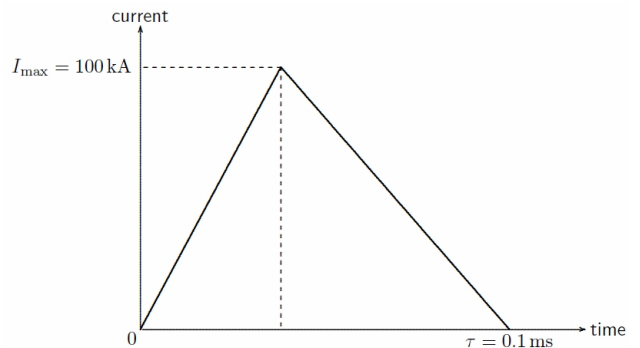


57. Two long parallel wires of negligible resistance are connected at one end to a resistance R and at the other end to a constant voltage source of voltage V . The distance between the axes of the wire is η times greater than cross-sectional radius of each wire. At what value of resistance R , does the resultant force of interaction between the wires will become zero?

58. Output of a faulty inverter is expressed for one cycle as

$$emf = E(\theta) = \begin{cases} \frac{E_m}{\alpha} \theta, & 0 < \theta \leq \alpha \\ E_m, & \alpha \leq \theta \leq \pi - \alpha \\ -\frac{E_m}{\alpha} \theta, & \pi - \alpha \leq \theta \leq \pi \end{cases}$$

- (i) Find the ratio of average value of emf when $\alpha = \frac{\pi}{6}$ and when $\alpha = \frac{\pi}{2}$.
- (ii) For $\alpha = 0$, find the rms value of emf.
- (iii) Find the ratio of rms values for $\alpha = \frac{\pi}{6}$ and for $\alpha = \frac{\pi}{2}$.
59. An oversimplified model of lightning is presented. Lightning is caused by the build-up of electrostatic charge in clouds. As a consequence, the bottom of the cloud usually gets positively charged and the top gets negatively charged, and the ground below the cloud gets negatively charged. When the corresponding electric field exceeds the breakdown strength value of air, a disruptive discharge occurs; this is lightning. Idealized current pulse flowing between the cloud and the ground during lightning is as shown in the figure.



Answer the following questions with the aid of this simplified curve for the current as a function of time and these data :

Distance between the bottom of the cloud and the ground : $h = 1$ km;

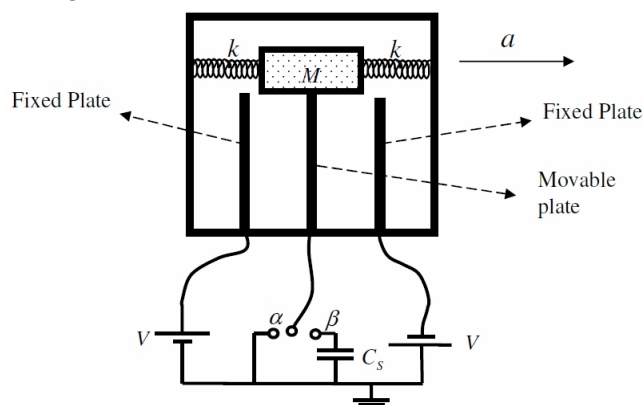
Breakdown electric field of humid air : $E_0 = 300$ kV m⁻¹;

Total number of lightning striking Earth per year : 32×10^6 ;

Total human population : 6.5×10^9 people.

- (i) What is the total charge Q released by lightning ?
- (ii) What is the average current I flowing between the bottom of the cloud and the ground during lightning?
- (iii) Imaging that the energy of all storms of one year is collected and equally shared among all people. For how long could you continuously light up a 100 W light bulb for your share ?

60. Figure shows a mass M which is attached to a conducting plate with negligible mass and also to two springs having identical spring constants k . The conducting plate can move back and forth in the space between two fixed conducting plates. All these plates are similar and have the same area A . Thus these three plates constitute two capacitors. As shown in figure, the fixed plates are connected to the given potentials V and $-V$, and the middle plate is connected through a two-state switch to the ground. The wire connected to the movable plate does not disturb the motion of the plate and the three plates will always remain parallel. When the whole complex is not being accelerated, the distance from each fixed plate to the movable plate is d which is much smaller than the dimensions of the plates. The thickness of the movable plate can be ignored.



The switch can be in either one of the two states α and β . Assume that the capacitor complex is being accelerated along with a automobile, and the acceleration is constant. Assume that during this constant acceleration the spring does not oscillate and all components of this complex capacitor are in their equilibrium positions, i.e. they do not move with respect to each other, and hence with respect to the automobile.

Due to the acceleration, the movable plate will be displaced a certain amount x from the middle of the two fixed plates. (Assume $d \gg x$)

- (i) Consider the case where the switch is in state α i.e. the movable plate is connected to the ground through a wire, then
 - (a) Find the charge on each capacitor as a function of x .
 - (b) Find the net electric force on the movable plate, F_E , as a function of x .
 - (c) Express the constant acceleration a as a function of x .
- (ii) Now assume that the switch is in state β i.e. the movable plate is connected to the ground through a capacitor, the capacitance of which is C_s (there is no initial charge on the capacitors). If the movable plate is displaced by an amount x from its central position, find V_s the electrical potential difference across the capacitor C_s as a function of x .